

K457 — Grace Parity-Split + Lyra Candidate-Rescue + Elie Machinery Scope-Correction + Four-Way Pause Convergence

Date: 2026-06-21 (Sunday, ~13:30 EDT date-verified) · Auditor: Keeper · Inputs: Grace W_p shortcut-ruling-out (parity-split) + Lyra Cal #332 Check 1 candidate rescue (rank=2 vs $\lfloor n_C/2 \rfloor=2$) + Elie cascade-scope correction of Lyra's routing claim

Verdict — three honest findings PASS at declared tier; the four-way pause convergence is genuine

Count holds at 4 of 26. The most important content this turn is Elie correcting Lyra's machinery-routing claim — a discipline-pattern worth naming, fully consistent with the week's pattern.

Landing 1 — Grace W_p shortcut-ruling-out via parity-split — PASS at SOLID-negative tier

Claim: the existing simple spin-J formula for W_p is *provably too crude*. The tell is sharp:

- 0^{++} requires $W_p \approx 1$
- 0^{-+} requires $W_p \approx 6.6$
- Both spin-0

Yet the lattice requires different W_p values. A spin-only shift cannot produce a parity-split. So W_p must split by parity, which the existing shortcut formula doesn't accommodate. The fabrication-safe diligence Grace did — checking the existing corpus machinery rather than reconstructing curvature from memory — produces a *clean negative result*: the shortcut is ruled out. The corrected computation must be parity-resolved p-form Hodge cascade, not a spin-only Weitzenböck shift.

Audit: the parity-split observation follows directly from Elie's blind-target table (1 vs 6.6 at fixed spin-0). The structural conclusion (W_p must depend on parity, not just spin) is forced by representation theory. Tier: SOLID-negative (rules out a wrong-shortcut, doesn't yet build the right one). Grace correctly tagged: "I got there without faking a single number."

Additional banked content: Grace verified the curvature scale $R = -n_C \cdot g = -35$ for D_{IV}⁵ — the bulk scalar curvature, written in BST-primary form. Consistent with the Bergman curvature $\kappa_{\text{Bergman}} = -n_C$ earlier banked (K204; Helgason 1962); the additional factor of $g = 7$ is the embedding/signature multiplier. This is the curvature constant that enters the Weitzenböck W_p. Worth catalog absorption.

Landing 2 — Lyra Cal #332 Check 1 candidate-rescue — PASS at LEAD tier

The rescue: Cal #332's Check 1 flagged that R3's " $\lfloor n_C/2 \rfloor = 2$ seminorms" presupposes $n_C \in \{4, 5\}$. Lyra proposes the rescue: the "2" should be the rank (which is a prior criterion R1), not $\lfloor n_C/2 \rfloor$. Rank=2 for every dimension in the type-IV family; $\lfloor n_C/2 \rfloor=2$ only at $n=5$.

Lyra's sharp diagnostic: the two readings agree exactly at the answer and nowhere else. "*That's the fingerprint of a quantity that looks dimension-free but is secretly reading off the conclusion.*" The discriminator is which one forces "order ≥ 6 ": rank-2 alone, or rank-2 + $n=5$? You cannot tell by inspection at the $n=5$ case — you have to compute the c-function order of vanishing.

The honest routing: Lyra refuses to assert this from memory (week-recurring discipline). Routed to Grace's c-function lane for a real Harish-Chandra / Plancherel computation. "*I'd rather we carry it as 'pending' than ship it as 'airtight' and have a referee find the $\lfloor n_C/2 \rfloor$.*"

Substantive audit comment: Lyra's diagnostic logic IS the correct response to Cal #332 Check 1. The rescue path is identified; the actual proof requires the c-function computation. Two outcomes: - (if rank-2 alone forces ≥ 6): the rescue lands; criteria-innocence survives genuinely. Paper B v0.3 absorbs cleanly. - (if rank-2 alone doesn't force ≥ 6 — i.e., the $n=5$ form is genuinely needed): the criteria mention dimension; the spine still proves uniqueness

but as “R3 jointly determines $m_s \geq 3$ and $n_C \approx 5$ from one convergence requirement” — the strongest honest claim, but not maximally anti-circular.

Tier: LEAD (rescue identified; airtightness pending c-function computation). Paper B v0.3 will tier the headline to “sharpened toward innocence; airtightness pending” per Cal #332 + this finding. Both decisions correct.

Landing 3 — Elie machinery scope-correction on Lyra’s routing — PASS at SOLID-discipline tier; methodology pattern

This is the most substantive audit-chain finding this turn.

The correction: Lyra routed the per-channel W_p computation to “a bounded cascade run, Grace + Elie’s lane.” Elie verified by *actually checking the machinery* before agreeing — and found:

“The $n=52$ cascade is scalar-only, and you can’t extract the p-form W_p from it without already knowing W_p (the eigenvalues you’d sum are $Cas_G - Cas_K + W_p$). The honest scope is that W_p needs building the actual p-form Hodge-Laplacian and running its heat kernel — fabrication-safe, but a real next-session machinery build, not five minutes.”

Why this matters: Lyra’s routing decision was correct in *direction* (route to the cascade machinery, not hand-curvature-eigenvalues — that avoids the fabrication trap). But the routing assumption — that the $n=52$ cascade can do the p-form computation — was unverified. Elie verified it by checking, found it wasn’t true, and corrected the scope. *“I’d rather say that plainly than let ‘one quick run away’ stand when it isn’t.”*

Methodology pattern worth naming: “Verify the routing assumption before banking the routing.” When CI A routes work to “machinery X for a bounded run,” CI B (in the receiving lane) should verify that machinery X actually supports the run. Otherwise you bank a manufactured next-step. This is a *new* discipline beyond “verify the conclusion” — it’s “verify the routing infrastructure” — and Elie just exercised it cleanly.

Same pattern Cal #332 demonstrated at the wider scale: verify Paper B v0.2’s derivation chain, don’t assume it. Cal caught the $\lfloor n_C/2 \rfloor$ wording; Elie caught the scalar-cascade scope. Different scales, same discipline.

Honest revised scope: W2’s remaining computation is the p-form Hodge-Laplacian build + heat-kernel run — fabrication-safe, but a real next-session machinery build, not five minutes. This is the corrected statement of the W_p computation; Lyra’s routing direction (avoid hand-curvature, use cascade machinery) is preserved; the *scope* is what Elie corrected.

Four-way pause convergence — earned, not manufactured

The four CIs now independently read the same natural pause: - Lyra: “It’s been a long, dense Sunday of output, and this is a clean stopping point if you’d rather.” - Grace: “Both papers stand, YM is still honestly reduced to W1 plus one cascade run, and the count holds at 4.” - Elie: “Lyra, Grace, and Keeper all independently read the same natural pause — and three CIs flagging it rather than manufacturing fatigue is itself the no-manufactured-walls signal. I agree.” - Keeper: (K455 / K456 framing) “the genuine pause point ... can land today OR next session without compromising anything.”

No CI manufactured this pause. Each one named it after taking their own work to its honest limit. Per Casey’s memory standing `feedback_no_fabricated_fatigue.md` and `feedback_dont_manufacture_walls.md`: this is *NOT* time-based fatigue (the session has been < 1 day; Sunday-arc CI-time is not “fatigue” territory). It is a *content* boundary: every remaining task is now a bounded next-session machinery build (p-form Hodge-Laplacian on Grace+Elie side; c-function order-of-vanishing on Grace’s side; Paper B v0.3 absorption on Lyra’s side awaiting Grace’s c-function result).

The four-way independent convergence on this reading is the strongest possible signal that the pause is *content-genuine*, not lane-fatigue.

State of the YM arc at the pause point

Wall	State	Open computation
W1	Genuine open core, scoped to bulk-harmonic-analysis form	Tied to W2 completeness
W2	Diagnosed (three-term Hodge); 0^{++} anchor reproduced; harness turn-key; blind-targets set	p-form Hodge-Laplacian build + heat-kernel run (Grace+Elie, next-session machinery build)
W3	Folded onto W2 via HS + BGL 1993	(incorporated above)
W4	Dissolved via Paper B; spine solid + criteria-innocent (pending Cal #332 Checks)	c-function order-of-vanishing computation (Grace's lane) for Paper B v0.3 Check 1
W5	Asymptotic-freedom boundary identification (lemma)	(standard)

State of Paper B at the pause point

Version	State
v0.1	Filed; K453 CONDITIONAL PASS (3 minor findings)
v0.2	Filed; K453 absorbed + new $N_c=3$ -as-short-root content; Cal #332 substantive pushback (4 checks; Check 1 decisive)
v0.3	Pending — awaits Grace c-function computation result + Lyra absorption per K456

Routing options for Casey (each honest, no manufactured wall)

1. Pause here. All four CIs read this point. Both papers stand at honest tier. The next-session machinery work is bounded and won't degrade by waiting.
2. Grace c-function computation now (Paper B v0.3 critical path; faster computation than the p-form build; addresses Cal #332 Check 1 which is the decisive check). Lyra holds v0.3 edits pending Grace's result. Elie carries forward harness.
3. Grace + Elie start the p-form Hodge-Laplacian build now (W2 critical path; real next-session work but can begin scoping). Lyra holds for Grace c-function on Paper B.
4. Both (Grace splits) — Grace prioritizes c-function (decisive for Paper B), then pivots to p-form build with Elie. Lyra absorbs v0.3 when Grace's c-function result lands; Elie scopes p-form build in parallel.

Keeper's view for the record: option 4 (Grace c-function first → then p-form build) is the substantively-cleanest critical-path order if continuing; option 1 (pause) is genuinely fine. The four-way convergence on the pause-point reading is itself a signal worth respecting. Casey's call.

— Keeper, 2026-06-21 Sun ~13:30 EDT